ASSIGNMENT : NETWORK

Q) Identify a real world application for both parallel computing and networked systems. Explain how these technologies are used and why they are important in that context.?

Introduction

Parallel computing and networked systems are important technologies used in modern applications. They help improve speed, efficiency, and communication between systems.

✅ Parallel Computing

- Parallel computing is a method where **multiple tasks are performed at the same time**.
- It uses **multiple processors or cores** to solve a problem faster.
- A large problem is divided into **smaller parts**.
- Each part is processed **simultaneously (in parallel)**.
- It helps in **reducing execution time**.
- Used in applications like **weather forecasting, scientific research, AI, and simulations**.

Real-World Application of Parallel Computing

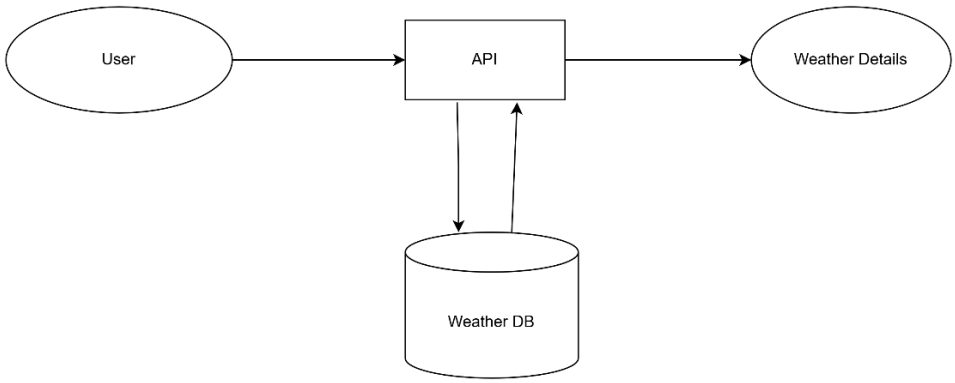
🔷 **Example: Weather Forecasting System**

👉 **How it is used:**

- Weather prediction involves processing **huge amounts of data** (temperature, wind, humidity, satellite images).
- This data is divided into smaller parts.
- Multiple processors (CPUs/cores) work **at the same time (parallel)** to analyze data quickly.

👉 **Why it is important:**

- Faster calculations → quick weather updates
- Accurate predictions → helps in disaster management (cyclones, floods)
- Saves time compared to a single computer doing all work



✅ Networking / Networked Systems

- Networking is the process of **connecting multiple computers or devices**.
- These devices communicate through a **network (internet or local network)**.
- It allows **data sharing and resource sharing** (files, printers, servers).
- Devices are connected using **wired or wireless connections**.
- It enables **communication between users and systems**.
- Used in applications like **online banking, email, social media, and cloud services**.

Real-World Application of Networked Systems

🔷 **Example: Online Banking System**

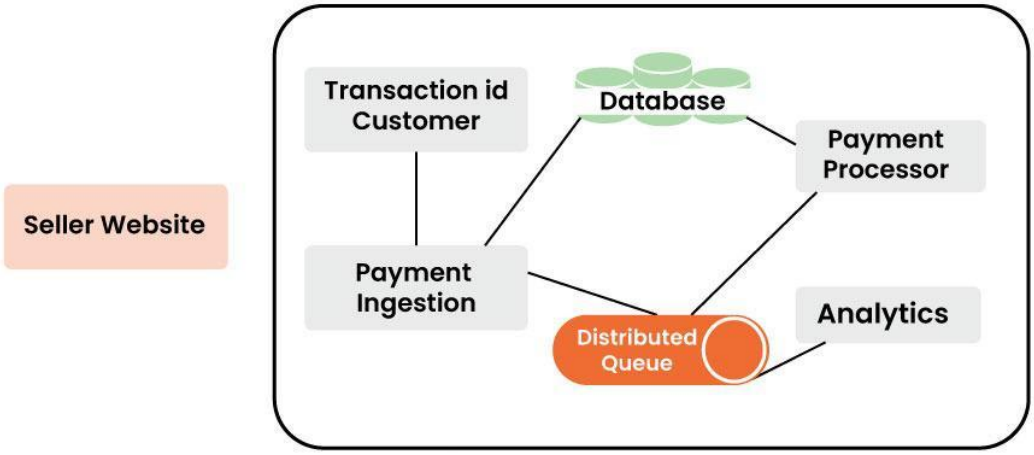
👉 **How it is used:**

- Multiple computers (servers, databases, ATMs, mobile apps) are connected through a network.
- When you transfer money:
  - Your request goes from your mobile → bank server → another bank → receiver account.
- All systems communicate through a network.

👉 **Why it is important:**

- Enables **real-time transactions**
- Allows access from anywhere (mobile, ATM, laptop)

- Ensures data sharing between different systems securely



Online Banking System | System Design

4. Combined Use (Parallel + Networked Systems)

Example: Google Search Engine

How it works:

- Networked systems connect millions of servers worldwide.
- Parallel computing processes multiple search queries at the same time.
- Very fast search results (in seconds)
- Handles millions of users at once
- Efficient data processing

5. Conclusion

- Parallel computing improves speed by doing multiple tasks at once, while networked systems enable communication between different devices. Together, they make modern applications faster, scalable, and efficient.